RACE
Regular Analysis Through Continuous Exercise

Trigonometric Ratios and Identities: RACE-01

Single Correct Question

1. The radius of a circle is 30 cm. if the length of the chord of the arc is 30 cm. then the length of smaller arc of the circle is

(A) 5πcm (B) 10πcm (C) 15πcm (D) 20πcm

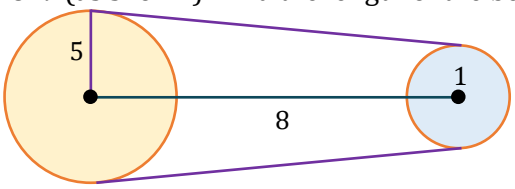
MTR131

2. A man is running around a regular hexagonal field of side length $6m$, so that he always at a distance 3 metre from the nearest boundary. The length of path travelled by him in one round is –

(A) $36 + 3\pi$ (B) $36 + 2\pi$ (C) $36 + 6\pi$ (D) $18 + 3\pi$

MTR132

3. A belt is tied up across two circular pulleys of radii $5m$ and $1m$ respectively whose centres are separated at a distance $8m$. (as shown). Find the length of the belt required.



(A) $\frac{22\pi}{3}+8\sqrt{3}$ (B) $6\pi+8\sqrt{3}$ (C) $\frac{22\pi}{3}+4\sqrt{3}$ (D) $6\pi+4\sqrt{3}$

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MTR133

4. If $\sin\theta + \text{cosec}\theta = 2$, then the value of $\sin^8\theta + \text{cosec}^8\theta$, is

(A) 2^8 (B) 2^4 (C) 2 (D) can't be determined

MTR134

5. The value of $\text{cosec}^6\alpha - \cot^6\alpha - 3\text{cosec}^2\alpha\cot^2\alpha$, is

(A) 1 (B) 2 (C) 3 (D) $\frac{1}{2}$

MTR135

Multiple Correct Question

6. If θ lies in first quadrant and $\cot\theta=\frac{4}{3}$, then

(A) $\sin\theta=\frac{4}{5}$ (B) $\sin\theta=\frac{3}{5}$ (C) $\cos\theta=\frac{4}{5}$ (D) $\text{cosec}\theta=\frac{5}{3}$

MTR136

7. Which of the following is/are correct

(A) $(\sin A + \cos A)(1 - \sin A\cos A) = \sin^3 A + \cos^3 A$.

(B) $\cos^6 A + \sin^6 A = 1 - 3\sin^2 A\cos^2 A$.

(C) $\sqrt{\frac{1-\sin A}{1+\sin A}}=\sec A-\tan A\ (-90^\circ < A < 90^\circ)$

(D) $\frac{\tan A}{1-\cot A}+\frac{\cot A}{1-\tan A}=\sec A\text{cosec}A+1$

MTR137

For More Join: @IITJEE_Advanced

8. If $\sin\theta = \frac{x^2-y^2}{x^2+y^2}$, where $\theta \in (0,90°)$. Then -

(A) $\cos\theta = \frac{2xy}{x^2+y^2}$ (B) $\cos\theta = \frac{2xy}{x^2-y^2}$ (C) $\cot\theta = \frac{2xy}{x^2+y^2}$ (D) $\cot\theta = \frac{2xy}{x^2-y^2}$

MTR138

Subjective Type Question

9. Convert the following measurement into radians :

(a) 25° 30' 30" (b) 10° 42' 30" (c) 9° 18' 42"

MTR139

10. Prove that :

(i) $\sec^4 A - \sec^2 A = \tan^4 A + \tan^2 A$.

(ii) $\tan^2 A - \sin^2 A = \sin^4 A \sec^2 A = \tan^2 A . \sin^2 A$

(iii) $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = 2$

MTR140

(iv) $\frac{1}{\operatorname{cosec} A - \cot A} - \frac{1}{\sin A} = \frac{1}{\sin A} - \frac{1}{\operatorname{cosec} A + \cot A}$

(v) $\frac{\cot A . \cos A}{\cot A + \cos A} = \frac{\cot A - \cos A}{\cot A . \cos A}$

(vi) $\sin^8 A - \cos^8 A = (\sin^2 A - \cos^2 A)(1 - 2\sin^2 A \cos^2 A)$

MTR141

(vii) $\frac{\tan A + \sec A - 1}{\tan A - \sec A + 1} = \frac{1 + \sin A}{\cos A}$

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(viii) $2\sec^2\alpha - \sec^4\alpha - 2\operatorname{cosec}^2\alpha + \operatorname{cosec}^4\alpha = \cot^4\alpha - \tan^4\alpha$.

MTR142

ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	B	C	A	C	A	B,C,D	A,B,C,D	A,D

9. (a) $\left(\frac{3061}{21600}\pi\right)$; (b) $\left(\frac{257}{4320}\pi\right)$; (c) $\frac{5587}{108000}\pi$

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Trigonometric Ratios and Identities: RACE-02

Single Correct Question

1. The value of $\cos570^\circ \sin510^\circ + \sin(-330^\circ)\cos(-390^\circ)$, is equal to

(A) 1 (B) 0 (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2}$

MTR143

2. The value of $\tan\frac{11\pi}{3} - 2\sin\frac{9\pi}{3} - \frac{3}{4}\operatorname{cosec}^2\frac{\pi}{4} + 4\cos^2\frac{17\pi}{6}$, is equal to

(A) $\frac{3+2\sqrt{3}}{2}$ (B) $3+2\sqrt{3}$ (C) $3-2\sqrt{3}$ (D) $\frac{3-2\sqrt{3}}{2}$

MTR144

Multiple Correct Question

3. If $\sin\theta = \frac{4}{5}$ then which of the following may be correct-

(A) $\cos\theta = \frac{3}{5}$ (B) $\cos\theta = \frac{-3}{5}$ (C) $\cot\theta = \frac{-4}{3}$ (D) $\cot\theta = \frac{4}{3}$

MTR145

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4. If $3\sec^4\theta + 8 = 10\sec^2\theta$, then the values of $\tan\theta$ can be-

(A) 1 (B) –1 (C) $\frac{-1}{\sqrt{3}}$ (D) $\sqrt{3}$

MTR146

5. If $\tan\theta = \frac{2x\ \ x+1}{2x+1}$, then:

(A) $\sin\theta = \pm\frac{2x(x+1)}{2x^2+2x+1}$ (B) $\sin\theta = \pm\frac{2x+1}{2x^2+2x+1}$

(C) $\cos\theta = \pm\frac{2x(x+1)}{2x^2+2x+1}$ (D) $\cos\theta = \pm\frac{2x+1}{2x^2+2x+1}$

MTR147

6. If $\tan^2\theta + \sec\theta = 5$, then value of $\cos\theta$ can be-

(A) $-\frac{1}{2}$ (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $-\frac{1}{3}$

MTR148

Integer Type Question

7. If $\tan\theta = \frac{1}{\sqrt{7}}$, find the value of $4\left(\frac{\operatorname{cosec}^2\theta - \sec^2\theta}{\operatorname{cosec}^2\theta + \sec^2\theta}\right)$, is

MTR149

For More Join: @IITJEE_Advanced

8. If $\tan\theta + \sec\theta = 1.5$, then value of $13\sin\theta$, is where $0 < \theta < \frac{\pi}{2}$

MTR150

9. If $\cos\theta = -\frac{1}{2}$ and $\pi < \theta < \frac{3\pi}{2}$, then find the value of $4\tan^2\theta - 3\text{cosec}^2\theta$.

MTR151

Subjective Type Question

10. **Evaluate:**

(a) $\sin 420° \cos 390° + \cos(-300°) \sin(-330°)$

MTR152

(b) $\tan 225° \cot 405° + \tan 765° \cot 675°$

MTR153

11. Express the following quantities in terms of the ratios of a positive angle, which is less than 45°

(a) $\sin(-65°)$ (b) $\cos(-928°)$ (c) $\tan 1145°$ (d) $\cot(-1054°)$

MTR154

12. What is the sign of $\sin A + \cos A$ for the following values of A ?

(a) 140° (b) –356° (c)–1125°.

MTR155

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	B	D	A,B	A,B,C	A,D	B,D	3	5	8

10. (a) 1; (b) 0

11. (a) –cos(25°); (b) –cos(28°); (c) cot 25°; (d) cot(26°)

12. (a) negative; (b) positive; (c) zero

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Trigonometric Ratios and Identities: RACE-03

Single Correct Type Question:

1. The value of $\sin 105° + \cos 105° =$

(A) 1 (B) 0 (C) $\frac{3}{2}$ (D) $\frac{1}{\sqrt{2}}$

MTR156

2. The value of $\tan\left(\frac{\pi}{4}+\theta\right)\times\tan\left(\frac{3\pi}{4}+\theta\right)$, (wherever defined) equals

(A) 1 (B) $\frac{1}{2}$ (C) $\sqrt{3}$ (D) –1

MTR157

3. The number of solutions of the equation $\sin(\pi x) + x^2 + 1 = 2x$, is

(A) 0 (B) 1 (C) 2 (D) 3

MTR158

4. If $\tan\alpha = \frac{5}{6}$ and $\tan\beta = \frac{1}{11}$, then $\alpha+\beta =$

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

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MTR159

Multiple Correct Type Question

5. Which of the following is/are correct

(A) The equation $\sin\theta = x + \frac{1}{x}$ is impossible if x be real.

(B) The equation $\sec^2\theta = \frac{4xy}{(x+y)^2}$ is only possible when $x = y$.

(C) The number of real solutions of the equation $\sin(e^x) = 2^x + 2^{-x}$ is 0

(D) The number of real solutions of the equation $\sin(e^x) = 2^x + 2^{-x}$ is 2

MTR160

6. If $\sin A = \frac{3}{5}$ and $\cos B = \frac{9}{41}$, $0 < A, B < \frac{\pi}{2}$, then

(A) $\sin(A+B) = \frac{187}{205}$ (B) $\sin(A-B) = \frac{-133}{205}$

(C) $\cos(A+B) = \frac{-84}{205}$ (D) $\cos(A-B) = \frac{156}{205}$

MTR161

For More Join: @IITJEE_Advanced

7. If $x + y = 45°$, which of the following is/are correct-

(A) $(1 + \tan x)(1 + \tan y) = 2$ (B) $(1 + \tan x)(1 + \tan y) = 1$

(C) $(\cot x - 1)(\cot y - 1) = 2$ (D) $(\cot x - 1)(\cot y - 1) = 1$

MTR162

8. If $\tan A = \frac{1}{2}$ and $\tan B = \frac{1}{3}$, then

(A) $\tan(2A + B) = \frac{9}{13}$ (B) $\tan(2A + B) = 3$

(C) $\tan(2A - B) = \frac{9}{13}$ (D) $\tan(2A - B) = 3$

MTR163

Subjective Type Question

9. Draw graphs of

(a) $y = \sin 2x$ (b) $y = 2\cos 3x$ (c) $y = 4\tan x$

MTR164

10. Prove that

(a) $\sin(45° + A)\cos(45° - B) + \cos(45° + A)\sin(45° - B) = \cos(A - B)$.

(b) $\cos(\alpha + \beta)\cos\gamma - \cos(\beta + \gamma)\cos\alpha = \sin\beta \sin(\gamma - \alpha)$

(c) $\sin(n + 1)A \sin(n - 1)A + \cos(n + 1)A \cos(n - 1)A = \cos 2A$

MTR165

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(d) $1 + \tan A \tan\frac{A}{2} = \tan A \cot\frac{A}{2} - 1 = \sec A$

(e) $\tan 3A \tan 2A \tan A = \tan 3A - \tan 2A - \tan A$.

MTR166

ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	D	D	C	B	A,B,C	A,B,C,D	A,C	B,C

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Trigonometric Ratios and Identities: RACE-04

Subjective Type Question

1. Simplify $\frac{\sin 75^\circ - \sin 15^\circ}{\cos 75^\circ + \cos 15^\circ}$

MTR167

2. Simplify $2\sin 54^\circ \sin 66^\circ$

MTR168

3. $\cos(36^\circ - A)\cos(36^\circ + A) + \cos(54^\circ + A)\cos(54^\circ - A) = \cos 2A$.

MTR169

4. $\sin(45^\circ + A)\sin(45^\circ - A) = \frac{1}{2}\cos 2A$

MTR170

5. $\sin(\beta - \gamma)\cos(\alpha - \delta) + \sin(\gamma - \alpha)\cos(\beta - \delta) + \sin(\alpha - \beta)\cos(\gamma - \delta) = 0$.

MTR171

6. $2\cos\frac{\pi}{13}\cos\frac{9\pi}{13} + \cos\frac{3\pi}{13} + \cos\frac{5\pi}{13} = 0$

MTR172

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7. $\frac{\sin 8\theta\cos\theta - \sin 6\theta\cos 3\theta}{\cos 2\theta\cos\theta - \sin 3\theta\sin 4\theta} = \tan 2\theta$

MTR173

8. $\frac{\sin 2A + \sin 2B}{\sin 2A - \sin 2B} = \frac{\tan(A+B)}{\tan(A-B)}$

MTR174

9. $\cos(A + B) + \sin(A - B) = 2\sin(45^\circ + A)\cos(45^\circ + B)$

MTR175

10. $\frac{\cos 3A - \cos A}{\sin 3A - \sin A} + \frac{\cos 2A - \cos 4A}{\sin 4A - \sin 2A} = \frac{\sin A}{\cos 2A\cos 3A}$

MTR176

11. $\frac{\cos 3\theta + 2\cos 5\theta + \cos 7\theta}{\cos\theta + 2\cos 3\theta + \cos 5\theta} = \cos 2\theta - \sin 2\theta\tan 3\theta$

MTR177

12. $\frac{\sin A + \sin 3A + \sin 5A + \sin 7A}{\cos A + \cos 3A + \cos 5A + \cos 7A} = \tan 4A$

MTR178

13. $\frac{\sin(A-C) + 2\sin A + \sin(A+C)}{\sin(B-C) + 2\sin B + \sin(B+C)} = \frac{\sin A}{\sin B}$

MTR179

For More Join: @IITJEE_Advanced

14. $\frac{\sin A+\sin B}{\sin A-\sin B}=\tan\frac{A+B}{2}\cot\frac{A-B}{2}$

MTR180

15. $\frac{\sin A-\sin B}{\cos B-\cos A}=\cot\frac{A+B}{2}$

MTR181

16. $\frac{\cos(A+B+C)+\cos(-A+B+C)+\cos(A-B+C)+\cos(A+B-C)}{\sin(A+B+C)+\sin(-A+B+C)-\sin(A-B+C)+\sin(A+B-C)}=\cot B$

MTR182

17. $\cos\left\{\theta+\left(n-\frac{3}{2}\right)\phi\right\}-\cos\left\{\theta+\left(n+\frac{3}{2}\right)\phi\right\}=2\sin\frac{3\phi}{2}.\sin(\theta+n\phi)$

MTR183

18. $\sin\frac{\theta}{2}\sin\frac{7\theta}{2}+\sin\frac{3\theta}{2}\sin\frac{11\theta}{2}=\sin2\theta\sin5\theta$

MTR184

19. $\cos2\theta\cos\frac{\theta}{2}-\cos3\theta\cos\frac{9\theta}{2}=\sin5\theta\sin\frac{5\theta}{2}$

MTR185

20. $\frac{2\sin(A-C)\cos C-\sin(A-2C)}{2\sin(B-C)\cos C-\sin(B-2C)}=\frac{\sin A}{\sin B}$

MTR186

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21. $\frac{\sin A\sin2A+\sin3A\sin6A+\sin4A\sin13A}{\sin A\cos2A+\sin3A\cos6A+\sin4A\cos13A}=\tan9A$

MTR187

22. $\frac{\cos2A\cos3A-\cos2A\cos7A+\cos A\cos10A}{\sin4A\sin3A-\sin2A\sin5A+\sin4A\sin7A}=\cot6A\cot5A$

MTR188

ANSWER KEY

1. $\frac{1}{\sqrt{3}}$ **2.** $\cos12°-\cos120°$

For More Join: @IITJEE_Advanced

Regular Analysis Through Continuous Exercise

Trigonometric Ratios and Identities: RACE-05

Subjective Type Question

Prove that :

1. $\frac{\sin 2A}{1+\cos 2A}=\tan A$ (Remember)

MTR189

2. $\frac{\sin 2A}{1-\cos 2A}=\cot A$ (Remember)

MTR190

3. $\tan A-\cot A=-2\cot 2A.$ (Remember)

MTR191

4. $\frac{1+\sin 2\theta+\cos 2\theta}{1+\sin 2\theta-\cos 2\theta}=\cot\theta$

MTR192

5. $\tan A+\cot A=2\operatorname{cosec}2A$

MTR193

6. $\frac{1-\cos A+\cos B-\cos(A+B)}{1+\cos A-\cos B-\cos(A+B)}=\tan\frac{A}{2}\cot\frac{B}{2}$

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MTR194

7. $\frac{\sec 8A-1}{\sec 4A-1}=\frac{\tan 8A}{\tan 2A}$

MTR195

8. $\cot(A+15^\circ)-\tan(A-15^\circ)=\frac{4\cos 2A}{1+2\sin 2A}$

MTR196

9. $\frac{\sin(n+1)A+2\sin nA+\sin(n-1)A}{\cos(n-1)A-\cos(n+1)A}=\cot\frac{A}{2}$

MTR197

10. $\cos^3 2\theta+3\cos 2\theta=4(\cos^6\theta-\sin^6\theta)$

MTR198

11. $\sec^2 A(1+\sec 2A)=2\sec 2A$

MTR199

12. $\operatorname{cosec}A-2\cot 2A\cos A=2\sin A$

MTR200

13. $\frac{2\cos 2^n\theta+1}{2\cos\theta+1}=(2\cos\theta-1)(2\cos 2\theta-1)(2\cos 2^2\theta-1)....(2\cos 2^{n-1}\theta-1)$

MTR201

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Trigonometric Ratios and Identities: RACE-06

Single Correct Question

1. If $\cos x+\sin x=a, \left(-\frac{\pi}{2}<x<-\frac{\pi}{4}\right)$, then $\cos 2x$ is equal to

(A) a^2 (B) $a\sqrt{(2-a)}$ (C) $a\sqrt{(2+a)}$ (D) $a\sqrt{\left(2-a^2\right)}$

MTR202

Subjective Type Question

2. If $\sin\theta=\frac{1}{2}$ and $\sin\phi=\frac{1}{3}$, find the values of $\sin(\theta+\phi)$ and $\sin(2\theta+2\phi)$.

MTR203

3. If $\cos\alpha=\frac{11}{61}$ and $\sin\beta=\frac{4}{5}$, find the values of $\sin^2\frac{\alpha-\beta}{2}$ and $\cos^2\frac{\alpha+\beta}{2}$, the angles α and β being positive acute angles.

MTR204

4. Find the values of

(a) $\sin 7\frac{1}{2}^\circ$ (b) $\cos 7\frac{1}{2}^\circ$ (c) $\tan 11\frac{1}{4}^\circ$

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MTR205

5. If $\sin\theta+\sin\phi=a$ *and* $\cos\theta+\cos\phi=b$, find the value of $\tan\frac{\theta-\phi}{2}$.

MTR206

Prove that :

6. (a) $\cot\theta\cot(60^\circ-\theta)\cot(60^\circ+\theta)=\cot 3\theta$

MTR207

(b) $\cos 5\theta=16\cos^5\theta-20\cos^3\theta+5\cos\theta$

MTR208

(c) $\cos 20^\circ\cos 40^\circ\cos 60^\circ\cos 80^\circ=\frac{1}{16}$

MTR209

7. (a) $\sin\alpha\sin(60^\circ-\alpha)\sin(60^\circ+\alpha)=\frac{1}{4}\sin 3\alpha$

MTR210

(b) $\cos\alpha\cos(60^\circ-\alpha)\cos(60^\circ+\alpha)=\frac{1}{4}\cos 3\alpha$

MTR211

(c) $\cot\alpha+\cot(60^\circ+\alpha)-\cot(60^\circ-\alpha)=3\cot 3\alpha$.

MTR212

For More Join: @IITJEE_Advanced

8. $\cos 6\alpha = 32\cos^6\alpha - 48\cos^4\alpha + 18\cos^2\alpha - 1.$

MTR213

9. $(\cos\alpha - \cos\beta)^2 + (\sin\alpha - \sin\beta)^2 = 4\sin^2\frac{\alpha-\beta}{2}$

MTR214

10. $\tan\left(45° + \frac{A}{2}\right) = \sqrt{\frac{1+\sin A}{1-\sin A}} = \sec A + \tan A$

MTR215

11. $\cos^2\alpha + \cos^2(\alpha + 120°) + \cos^2(\alpha - 120°) = \frac{3}{2}$

MTR216

12. $\sin^4\frac{\pi}{8} + \sin^4\frac{3\pi}{8} + \sin^4\frac{5\pi}{8} + \sin^4\frac{7\pi}{8} = \frac{3}{2}$

MTR217

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ANSWER KEY

1. (D)

2. $\frac{\pm 2\sqrt{2} \pm \sqrt{3}}{6}; \frac{\pm 7\sqrt{3} \pm 4\sqrt{2}}{18}$

3. $\frac{16}{305}; \frac{49}{305}$

4. (a) $\frac{\sqrt{4-\sqrt{2}-\sqrt{6}}}{2\sqrt{2}}$ (b) $\frac{\sqrt{4+\sqrt{2}+\sqrt{6}}}{2\sqrt{2}}$ (c) $\sqrt{4+2\sqrt{2}} - (\sqrt{2}+1)$

5. $\pm\sqrt{\frac{4-a^2-b^2}{a^2+b^2}}$

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Trigonometric Ratios and Identities: RACE-07

Subjective Type Question

1. Find the value of

(a) $\sin\frac{\pi}{10}+\sin\frac{13\pi}{10}$ (b) $\cos^2 48^\circ-\sin^2 12^\circ$

MTR218

2. $\sin\frac{\pi}{5}\sin\frac{2\pi}{5}\sin\frac{3\pi}{5}\sin\frac{4\pi}{5}$

MTR219

3. $\tan6^\circ\tan42^\circ\tan66^\circ\tan78^\circ$

MTR220

4. $\cos\frac{\pi}{15}\cos\frac{2\pi}{15}\cos\frac{3\pi}{15}\cos\frac{4\pi}{15}\cos\frac{5\pi}{15}\cos\frac{6\pi}{15}\cos\frac{7\pi}{15}$

MTR221

5. If $A=\cos\frac{\pi}{9}\cos\frac{4\pi}{9}\cos\frac{16\pi}{9}, B=\cos\frac{2\pi}{9}\cos\frac{8\pi}{9}\cos\frac{32\pi}{9}$, then value of $\left|\frac{1}{16A.B}\right|$ is

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MTR222

6. In any circle prove that the chord which subtends 108° at the centre is equal to the sum of the two chords which subtend angles of 36° and 60°.

MTR223

7. If $x+y+z=xyz$ prove that

(a) $\frac{3x-x^3}{1-3x^2}+\frac{3y-y^3}{1-3y^2}+\frac{3z-z^3}{1-3z^2}=\frac{3x-x^3}{1-3x^2}.\frac{3y-y^3}{1-3y^2}.\frac{3z-z^3}{1-3z^2}$

MTR224

(b) $x(1-y^2)(1-z^2)+y(1-z^2)(1-x^2)+z(1-x^2)(1-y^2)=4xyz.$

MTR225

For Question Number 8 to 15, consider $A+B+C=\pi$. Prove that

8. $\sin2A+\sin2B-\sin2C=4\cos A\cos B\sin C$

MTR226

9. $\cos2A+\cos2B-\cos2C=1-4\sin A\sin B\cos C$

MTR227

10. $\sin^2A+\sin^2B-\sin^2C=2\sin A\sin B\cos C$

MTR228

11. $\cos^2A+\cos^2B-\cos^2C=1-2\sin A\sin B\cos C$

MTR229

For More Join: @IITJEE_Advanced

12. $\sin^2\frac{A}{2}+\sin^2\frac{B}{2}-\sin^2\frac{C}{2}=1-2\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$

MTR230

13. $\sin(B+2C)+\sin(C+2A)+\sin(A+2B)=4\sin\frac{B-C}{2}\sin\frac{C-A}{2}\sin\frac{A-B}{2}$.

MTR231

14. $\frac{\sin 2A+\sin 2B+\sin 2C}{\sin A+\sin B+\sin C}=8\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

MTR232

15. $\sin(B+C-A)+\sin(C+A-B)+\sin(A+B-C)=4\sin A\sin B\sin C$

MTR233

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ANSWER KEY

1. (a) $-\frac{1}{2}$ (b) $\frac{\sqrt{5}+1}{8}$ **2.** $\frac{5}{16}$ **3.** (1) **4.** $\frac{1}{2^7}$

5. (4)

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RACE
Regular Analysis Through Continuous Exercise

Trigonometric Ratios and Identities: RACE-08

Single Correct Question

1. If $S=\cos^2\frac{\pi}{n}+\cos^2\frac{2\pi}{n}+...+\cos^2\frac{(n-1)\pi}{n}$, then S equals (where $n\geq 2, n\in N$)

(A) $\frac{n}{2(n+1)}$ (B) $\frac{1}{2(n-1)}$ (C) $\frac{n-2}{2}$ (D) $\frac{n}{2}$

MTR234

Integer Type Question

2. Find the minimum value of $4\sec^2 x+9\operatorname{cosec}^2 x$

MTR235

3. Evaluate $\sin\frac{\pi}{n}+\sin\frac{3\pi}{n}+\sin\frac{5\pi}{n}+.....$ to n terms

MTR236

Subjective Type Question

4. Find minimum value of

(i) $3\sin^2 x+27\operatorname{cosec}^2 x$ (ii) $27\sin^2 x+3\operatorname{cosec}^2 x$

MTR237

5. Find maximum and minimum value of $5\cos\theta+3\sin\left(\theta+\frac{\pi}{6}\right)$ for all real values of θ.

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MTR238

6. Find the minimum value of $\cos\theta+\cos 2\theta$ for all real values of θ.

MTR239

7. Find maximum and minimum value of $\cos^2\theta-6\sin\theta\cos\theta+3\sin^2\theta+2$.

MTR240

8. If $\alpha+\beta=90°$, then find the maximum value of $\sin\alpha.\sin\beta$.

MTR241

9. Find sum of the following series :

(a) $\cos\frac{\pi}{2n+1}+\cos\frac{3\pi}{2n+1}+\cos\frac{5\pi}{2n+1}+...$ up to n terms.

MTR242

(b) $\sin 2\alpha+\sin 3\alpha+\sin 4\alpha+...+\sin n\alpha, where (n+2)\alpha=2\pi$

MTR243

10. Prove that : $\cos\frac{\pi}{11}+\cos\frac{3\pi}{11}+\cos\frac{5\pi}{11}+\cos\frac{7\pi}{11}+\cos\frac{9\pi}{11}=\frac{1}{2}$

MTR244

ANSWER KEY

1.	(C)	**2.**	(25)	**3.**	(0)	**4.**	(i) 30 (ii) 18
5.	(7 & –7)	**6.**	$-\frac{9}{8}$	**7.**	$4+\sqrt{10}$, $4-\sqrt{10}$	**8.**	$\frac{1}{2}$
9.	(a) $\frac{1}{2}$ (b) (0)						

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